

# CHIMERA-QRC: A Pre-Registered, Adversarially-Controlled Study of Quantum Reservoir Computing for S&P 500 Realized-Volatility Forecasting

Team EIGENNEXUS — C. Metzl, F. Eldibani, J. M. Aguiar Hualde · Track A (Financial Volatility) · GIC 2026 Phase 3

**Abstract.** We contribute a **pre-registered, adversarially-controlled evaluation instrument** for quantum reservoir computing — reproducible **offline in one command** (`cli.py verify`), with every hardware prediction bound to a prior commit by a **hash-preimage commitment** — and apply it to S&P 500 realized-volatility forecasting across **four devices from three QPU vendors** (IonQ, IQM, Rigetti). Two measurements stand on their own: an empirical **quantum-complexity metric** (the reservoir's MPS bond dimension is essentially full,  $\chi_{\text{eff}} \approx 2^{(n/2)}$ , marking the classical-simulability boundary) and a cross-platform **coherence-budget wall** (Fig. 3) whose readout-corrected fingerprint separates **coherent circuit routing from depolarizing/damping noise**, each bootstrapped regime verdict **9.7–23.9 $\sigma$**  beyond shot noise. Applied honestly, the instrument returns a **well-characterized negative**: at simulable scale ( $\leq 16$  qubits) the reservoir is *distinct but not more accurate* than the strongest fair baselines — **HAR-X**, size-matched ESN/RFF, under HAC-DM + Holm + MCS (**H0 refuted**) — robust across a second **Heisenberg** reservoir family (not Ising-specific) and a second (weather) domain, and our program **falsified four of its own pre-registered predictions** under controls (a same-session cross-seed control shows the  $n=8$  scrambling is *seed-0-instance-specific, not a size law*). **Desk takeaway:** vol-timing a simple RV forecast  $\approx$ halves the 2008 drawdown; quantum adds no economic value at this scale.

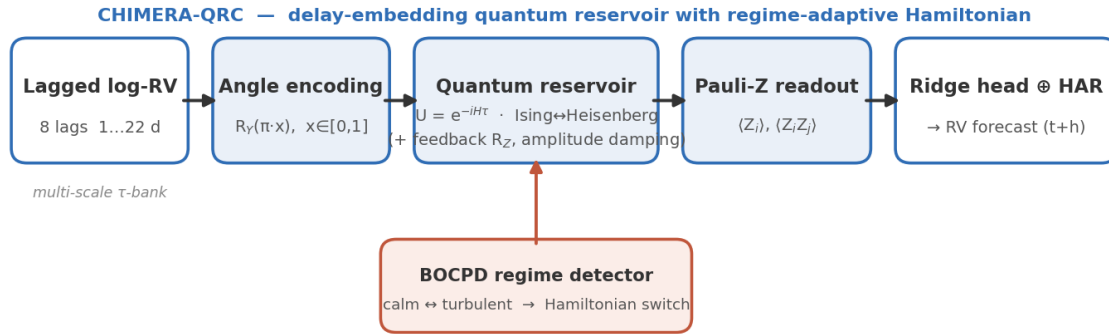
## 1. Focus, track selection, and problem framing

We target **Track A: financial volatility** — one-step-ahead forecasting of S&P 500 **realized variance (RV)** and the tracking of volatility **regime transitions**, the domain where JonesTrading's risk, allocation and derivatives pricing live. RV is long-memory, multi-scale, regime-switching and non-Gaussian — the structure reservoir computing exploits. Our central yardsticks are the Echo State Network (ESN, the classical analog of QRC) and the strongest econometric models (HAR — *Heterogeneous AutoRegressive*, Corsi 2009 — and, critically for Phase 3, **HAR-X**: HAR augmented with the same realized-measure features we feed the quantum reservoir). We also report GARCH/GJR-GARCH, AR(3), persistence, LSTM, and an RFF/RBF kernel, and we implement the challenge's common **MNIST** benchmark on the identical engine.

**This paper's honest thesis.** We pre-registered H0 and *attacked our own result* with fair controls: the dominant accuracy lever is *which features are encoded*, not quantum nonlinearity. What survives for the reservoir — kernel distinctness, lower seed variance, full-rank entanglement (classical-simulation hardness) — is *necessary, not sufficient* for advantage. **Vs. prior work.** The closest study (Li et al. 2025/26 — QRC for realized volatility) presents the approach as a competitive, noise-resilient *proof-of-concept*; we add the control-hardened, pre-registered evaluation it omits — the decisive **HAR-X** control, HAC-DM/Holm/MCS, and explicit capability and per-layer-noise audits — turning an encouraging proof-of-concept into a falsifiable result.

## 2. QRC architecture (Fig. 1)

CHIMERA is a delay-embedding quantum reservoir. Inputs are angle-encoded  $R_Y(\pi \cdot x)$ , one value per qubit, onto  $|0\dots 0\rangle$ ; the state evolves under a fixed transverse-field **Ising** Hamiltonian  $H = \sum_{\{i<j\}} J_{ij} Z_i Z_j + h_x \sum_i X_i$  ( $h_x=1$ ;  $J$  a **random  $\approx 50\%$ -connected** graph,  $\text{connectivity}=0.5$ ;  $U=\exp(-iH\tau)$ ,  $\tau=2$ ); single- and pairwise Pauli-Z expectations  $\langle Z_i \rangle$ ,  $\langle Z_i Z_j \rangle$  form the feature vector. **A ridge head consumes these features concatenated with a linear block of the same inputs (incl. HAR)**, so any reservoir gain is genuine nonlinearity *beyond* the linear span of identical information. The identical inputs feed the classical controls, isolating quantum-vs-classical. Four mechanisms extend the core (multi-scale  $\tau$ -bank; RZ measurement feedback; regime-adaptive Ising $\leftrightarrow$ Heisenberg via BOCPD; dissipation-as- feature); Phase 3 adds an **encoding-density / data-re-uploading path** (§5.2) and a **sparse/tensor backend** (§5.5). The engine is pure NumPy; an explicit PennyLane circuit reproduces it to  $\approx 3.9 \times 10^{-16}$  and compiles, at  $n=8$ , to **380 native gates** (220 two-qubit + 160 one-qubit, 20 Trotter layers) — consistent with the  $\approx 50\%$ -connected graph.



**Figure 1.** CHIMERA-QRC pipeline: lagged log-RV (and, in Axis B, realized-measure) inputs are angle-encoded onto qubits, evolved under a fixed random-coupled Ising Hamiltonian, and read out as single/pairwise Pauli-Z expectations into a ridge head fused with a linear block of the same inputs; a BOCPD detector selects the Hamiltonian for regime adaptivity.

### 3. Theoretical and analytical justification

The exploited quantum resources map to RV's structure: (a) **Hilbert-space dimensionality** —  $n$  qubits give  $2^n$  amplitudes and  $O(n^2)$  measured Pauli features at  $O(n^2)$  parameter cost; (b) **intrinsic nonlinearity** from fixed unitary evolution (no gradient training, no barren plateaus); (c) **delay-embedded memory** (the headline model's memory is its explicit lag window, not reservoir recurrence — a recurrent variant is probed in `v3_research`); (d) **Hamiltonian as inductive bias**. The central, falsifiable mechanism is **expressivity scaling**, which we measure directly via the geometric difference (Huang et al. 2021). The quantum kernel is poorly reproducible by a matched classical reservoir:  **$g(\text{ESN} \rightarrow \text{CHIMERA}) \approx 64$  vs  $\approx 3.7$  control** (`cli.py run kernel`); against per- $n$  *feature-matched* ESNs it is larger but **non-monotonic** ( $\approx 125/158/88$  at  $n=8/10/12$ ; fixed ridge — the qualitative separation, not the value, is the claim). At  $\leq 12$ – $16$  qubits the map stays classically simulable, so distinctness is **necessary, not sufficient** — which we now *measure*: over **30 seeded  $n=8$  instances**, per-instance entanglement is **uncorrelated** with forecast error ( $r=-0.19$ ,  $p=0.32$ ) and **0/30 beat HAR-X** on crisis *and* calm. Nor does instance quality **transfer** — crisis/calm rankings are unrelated (Spearman  $+0.01$ ,  $p=0.96$ ; the crisis-best instance ranks 19/30 on calm) — so best-of- $N$  seed selection is selection on noise. Detectable  $|r| \gtrsim 0.5$ , so we exclude a *strong* link, not a weak one (`expressivity_accuracy_findings.md`, `exploratory`).

### 4. Data modeling strategy

**Datasets (all bundled in-repo).** Oxford-Man `.SPX` 5-minute realized variance, 2000–2020, 5,029 supervised days incl. the 2008 GFC (Heber et al. 2009; via R packages `highfrequency/bvhar`); SPY daily OHLCV 2022–2026 (Garman–Klass proxy; retrieved via Massive.com API, bundled); real MNIST (Keras `.npz` mirror). **Preprocessing:** log-RV target; features min-max scaled to  $[0,1]$  on **train only**; per-model ridge penalty selected on a chronological validation tail; multi-seed ensembling. We verified the pipeline is leakage-free (all features lagged  $\geq 1$  day; train-only scaling). **Baselines:** HAR, **HAR-X**, GARCH(1,1), GJR-GARCH, AR(3), persistence, ESN (matched +  $4\times$ , recurrent), RFF kernel, LSTM. **Metrics:** RMSE(log-RV), QLIKE (Patton 2011), Mincer–Zarnowitz  $R^2$ ; significance by **Diebold–Mariano with a Newey–West HAC variance** (HLN-corrected), **Holm**-adjusted across the comparison family, and the Model Confidence Set (Hansen–Lunde–Nason 2011).

### 5. Phase-3 execution — concrete results

**Pre-registration (honesty).** Before running, we fixed the confirm/refute thresholds in `preregistration.py` (H0/H1/H4, from our Phase-2 §7). All outcomes — including negatives — are reported against them. Full numbers/wall-clock: `results/*_findings.md`.

**5.1 The input-bottleneck mechanism (pre-registered negative).** With a *fixed* univariate 8-lag encoder, extra qubits receive no new input:  $g(n)$  and effective feature-rank **saturate/decline** ( $n=8 \rightarrow 12$ :  $g$  133  $\rightarrow$  30 at  $N_{\text{sub}}=800$ ;  $D_{\text{eff}}$  1.8  $\rightarrow$  1.5). **H0 is refuted in this regime** — the empirical case *for* enriching the encoding.

**5.2 Encoding density (Axis B) — mechanism confirmed.** Feeding the extra qubits genuinely new realized-measure information (signed return/leverage, downside-semivariance share, jump share) restores the

lost structure: at  $n=10$ , informed vs idle qubits, kernel distinctness and rank jump ( $g\ 52 \rightarrow 158$ ,  $D_{\text{eff}}\ 1.5 \rightarrow 3.1$ ), and effective rank grows with  $n$  ( $D_{\text{eff}}\ 1.81 \rightarrow 3.14$ ,  $n=8 \rightarrow 12$ ). So the bottleneck is *fixable* — added qubits can carry new information. The decisive question is whether this converts into forecasting **accuracy**.

**5.3 The decisive, adversarially-controlled test (the honest headline).** We compare CHIMERA to the strongest fair baselines — **HAR-X** (the same rich features used *linearly*, no reservoir), a **true recurrent ESN**, and an **RFF kernel** — all sharing the identical information set and *nesting* HAR-X, so a quantum win requires nonlinearity beyond the linear span. With **8 seeds, HAC-DM, two windows, and Holm correction** (`cli.py run axisB_rig`):

| RMSE(log-RV)  | HAR-X         | recurrent ESN | RFF    | CHIMERA | best  |
|---------------|---------------|---------------|--------|---------|-------|
| crisis $n=8$  | <b>0.6255</b> | 0.6386        | 0.6302 | 0.6379  | HAR-X |
| crisis $n=10$ | 0.6034        | <b>0.5996</b> | 0.6047 | 0.6074  | ESN   |
| crisis $n=12$ | <b>0.5906</b> | 0.6023        | 0.5964 | 0.6031  | HAR-X |
| calm $n=10$   | <b>0.6244</b> | 0.6445        | 0.6264 | 0.6291  | HAR-X |

(plain HAR for reference: crisis 0.6290, calm 0.6454 — HAR-X's rich features cut RMSE sharply.) **HAR-X is best or co-best everywhere; CHIMERA never beats it** and is indistinguishable from the classical ESN/RFF after Holm (one raw loss to RFF at crisis  $n=8$ ,  $p=0.049$ ). **After Holm correction no comparison is significant in either direction, and the 95% MCS retains all four models ( $n=10$ )** — a statistical tie. The earlier "beats HAR" result was an artifact of a *feature-poor* HAR: the gain is the encoded realized measures (a known SHAR/HARQ effect), not quantum nonlinearity. **By our pre-registered criteria,  $H_0$  is refuted — we report this negative.** What honestly survives for the quantum reservoir: it is **competitive** (within  $\approx 0.8$ – $2.1\%$  RMSE of the best at every  $n$ ), with **lower per-seed dispersion than the recurrent ESN in 3 of 4 cells** ( $n=12$ : 0.009 vs 0.015) and **beats it on the calm window** (raw  $p=0.018$ ; n.s. after Holm). A **tuned, size-unconstrained ESN** (45-config search to 800 nodes) confirms the control was not crippled (0.6020 vs CHIMERA 0.6094, within noise of HAR-X); and **named canonical models** (SHAR, HAR-CJ, HARQ, HEAVY-RM; MSE + QLIKE DM) — **none beats HAR-X**, with CHIMERA tying best on RMSE and only a *raw, non-Holm* QLIKE/MZ edge (`cli.py run canonical, esn_tuning_robustness_findings.md`).

Rigorous Axis-B: HAR-X (rich linear) is best; quantum shows no significant edge (Holm n.s.)

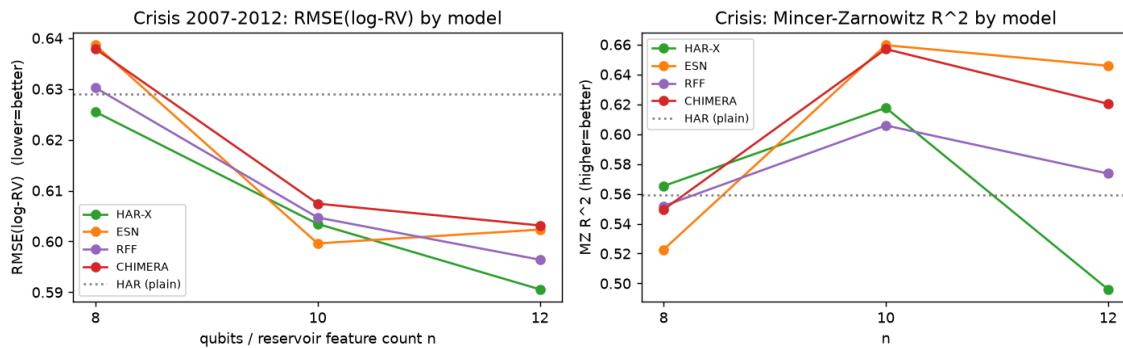


Figure 2. Rigorous Axis-B (8 seeds, crisis window). HAR-X (rich features, linear) is the best or co-best model at every  $n$ ; the quantum reservoir is competitive but shows no advantage that survives HAC-DM + Holm correction. Left: RMSE(log-RV). Right: Mincer–Zarnowitz  $R^2$ .

**5.4 Common MNIST benchmark + noise.** Same engine, pixels  $\rightarrow$  PCA( $n$ )  $\rightarrow$   $n$  qubits  $\rightarrow$  Pauli readout  $\rightarrow$  ridge. Accuracy **scales with qubits** ( $n=5/8/10/12$ : 0.632/0.798/0.831/0.859, 3 seeds;  **$n=15$  sparse-exact continues the trend**, 0.878 vs 0.852 for a paired  $n=12$  on a matched subset — closing the brief's 5/10/15 example) and **beats the linear-PCA baseline at every  $n$**  (real nonlinear lift), confirming **sufficient expressivity**; a matched ESN **ties or slightly exceeds CHIMERA** (within  $\approx 1\%$  for  $n \geq 8$ ; 2.9% at  $n=5$ ) — competitive, not dominant. **Noise:** the classifier is **invariant to depolarizing noise** (a uniform Bloch contraction that per-feature standardization removes exactly; accuracy identical across rates 0.05–0.30) and **robust to amplitude damping** ( $<0.5\%$  at 30%). **Honesty check** (`cli.py run noise_circuit`): a per-Trotter-layer density-matrix study shows the *converse* — noise interleaved with the evolution (per-layer single-qubit channels) is **not** removed by standardization (standardized error grows with rate, vs  $\approx 0$  for readout-only depolarizing): the invariance is a *readout* property, not circuit-level robustness; accumulated gate error is the real NISQ cost (§6).

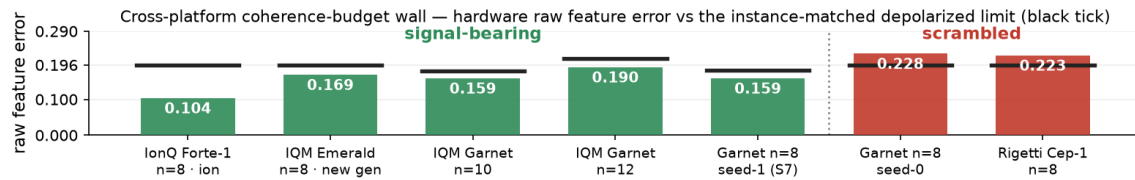
**5.5 Scaling frontier + quantum-complexity metric.** A sparse-exact backend (`expm_multiply`, no dense propagator; matches the dense engine to  $2.4 \times 10^{-14}$ ) reaches **n=16 exactly**. For the random  $\approx 50\%$ -connected reservoir we measure the **bond dimension**  $\chi_{\text{eff}}$  across a balanced cut: it is **essentially full at every n**,  $\chi_{\text{eff}} \approx 2^{(n/2)}$  ( $16 \rightarrow 255.9$  for  $n=8 \rightarrow 16$ ) — an exact MPS gets **zero compression** ( $S \approx 1.7\text{--}3.2$  nats); the reservoir admits **no low-bond-dimension (MPS/TEBD) shortcut**, exact cost stays exponential — the precondition any beyond-frontier advantage would need, though none appears at the scale we can test. **Capability + efficiency checks.** An information-processing-capacity probe (Dambre et al. 2012; `cli.py run capacity`) finds the quantum reservoir **not more** — slightly *less* — nonlinearly expressive than a matched RFF/ESN: no excess expressivity to exploit. A frontier check (`cli.py run frontier`) adds that  $g(n)$  does **not** widen toward  $n=16$  (it declines;  $D_{\text{eff}}/\text{rank}$  grow but a matched ESN keeps pace). **Robustness (supporting study).** *Second family:* a **Heisenberg (XXZ)** reservoir — equally kernel-distinct ( $g \approx 55$ ) — also fails to beat HAR-X (0.650 vs 0.642, 8 seeds, full-sample config), so the negative is **not Ising-specific** (`cli.py run second_family`). *Efficiency:* with input held fixed, a *smaller* QRC cannot substitute for a *larger* classical reservoir — quantum accuracy **saturates** while the classical curves improve; per feature the static maps are comparable, so the negative is **saturation, not per-feature inferiority** (weather RMSE °C: CHIMERA 0.85 at its qubit-range ceiling vs RFF 0.78 / recurrent ESN 0.71 at larger budgets). *Domain/architecture:* the *same* engine/protocol on chaotic **weather** (5 stations) and the autonomous **VPT** metric still show no advantage; the **recurrent** QRC is competitive-not-better — unitary evolution is non-dissipative, lacking the contraction behind ESN "generalized synchronization" (Ahmed-Tennie-Magri 2025). We then **fixed the mechanism**: engineered memory-qubit damping traces the pre-registered inverted-U, lifting autonomous VPT  $\approx +60\%$  to **parity** with the matched ESN — not better (`results/dissipative_qrc_findings.md`).

## 6. Quantum platform and resource planning

Simulator-first on qBraid: statevector  $\leq 12$  qubits, sparse/TN to  $\approx 16$ , GPU beyond. **Resource estimates** (gate-Trotter, 20 layers):  $n=8/10/12$  use **220/480/760 two-qubit + 160/200/240 one-qubit** gates, yielding **36/55/78 observables** — all Z-basis-diagonal, so **one S-shot set** ( $S \approx 2\text{--}8\text{k}$ ,  $\epsilon \approx 1/\sqrt{S}$ ) **reads every observable** (statevector build  $\leq 1$  min over this range).

**QPU validation** uses this gate-Trotter circuit on **IonQ / IQM / Rigetti** via qBraid — the random-sparse Ising needs *fewer* two-qubit gates than an all-to-all reservoir, easing NISQ mapping — with **ZNE + measurement mitigation** and a classical cross-check per run. The submission path is **executable now** (`cli.py run qsubmit`): on a simulator it (i) reproduces the engine to  $3.9 \times 10^{-16}$  (the per-run classical cross-check), (ii) characterizes the **shot budget** ( $\epsilon \approx 1/\sqrt{S}$ ; feature error  $0.046 \rightarrow 0.0066$  at  $S=256 \rightarrow 16\text{k}$ ; gate-Trotter(20) adds  $\approx 0.04$ ), and (iii) demonstrates **zero-noise extrapolation** under a depolarizing sweep. **Executed on real hardware:** the full mitigated protocol ran first on **Rigetti Cepheus-1-108q** (12 logged cloud jobs; bit order auto-detected; readout 2.3%/6.4%); after lattice routing, scale-1 already exceeds the coherence budget — the accumulated two-qubit-gate cost  $\$5.4$  predicts, measured on metal (ZNE marginal; a 4k-shot replication showed the beyond-limit regime stable under day-scale drift). The pre-registered **Garnet protocol** replicated the regime on a second vendor (raw 0.230 > our routing-free 0.171 forecast — **prediction (iii) confirmed**). Execution also surfaced a **platform-level finding**: negative-angle rotations are **silently lost server-side on the IonQ route** ( $\text{RY}(\pi)\text{RY}(-\pi) \rightarrow \text{net RY}(\pi)$ ), corrupting ZNE's negated folds; we hardened the emitter ( $\text{mod-}2\pi$ ) and validated the diagonal calibration on-device. The routing-free **IonQ Forte-1 campaign** then ran under the pre-committed manifest: a smoke gate measured a **2,000-gate/circuit ceiling** on the native route, so per abort rule it ran scale-1-only (500 shots — a disclosed, cost-forced deviation,  $\text{s.e.} \approx 0.004$ ). **Raw 0.104 — below the 0.196 depolarized limit** and every superconducting  $n=8$  number: a **signal-bearing hardware execution** of this reservoir (**prediction (ii) confirmed**; `results/qpu_hardware_findings.md`). A pre-registered **hardware scaling program** (`results/qpu_scaling_outlook.md`) then refuted our own statements on metal (Fig. 3): Garnet turns **signal-bearing at  $n=10/n=12$**  while the **same-session seed-0  $n=8$  anchor** stays scrambled — but a pre-registered cross-seed control (S7) found an **independent seed-1  $n=8$  instance signal-bearing** in the same session, so the  $n=8$  scrambling is **specific to the seed-0 instance, not a size law** (seed-1  $n=10$  also signal-bearing) — and the newer-generation **Emerald is signal-bearing at the very size Garnet scrambles**, reproduced in a **same-window two-chip pair**: generation-dependent at fixed size, temporally controlled. The program characterizes reservoir size, encoding density, shot budget and noise, with in one view. The criterion is

exactly  $\text{SNR} > 1$  — raw error is  $\text{mean}|F - F_{\text{exact}}|$ , the limit is  $\text{mean}|F_{\text{exact}}|$  — so the bar is **instance-matched** (signal magnitude is a property of the circuit, not of  $n$ ) and the test transfers to any quantum feature map:



**Figure 3.** Cross-platform coherence-budget wall: mean raw feature error vs the **instance-matched** depolarized limit  $\text{mean}|F_{\text{exact}}|$  (black ticks; 4k shots, IonQ 500). Five configs are signal-bearing (below the limit), two scrambled (Garnet  $n=8$  seed-0 0.228; Rigetti 0.223). The adjacent Garnet  $n=8$  seed-1 (0.159) vs seed-0 (0.228) pair — same chip and session — refutes a *size* law. Seed-0 limits 0.196/0.179/0.214 at  $n=8/10/12$ ; seed-1 carries its own 0.1806 — verdict unchanged (§6).

**Execution provenance (externally verifiable).** Every campaign ran against pre-committed predictions — the ten funded ones under a committed manifest (abort/decision rules fixed *before* each launch); every funded-campaign job embeds the repo commit hash and campaign tag in qBraid's timestamped records — a hash-preimage commitment that predictions predate data — and billing matched estimates to the half-credit (results/CREDIT\_BUDGET.md:  $\approx 64\text{k}$  credits). Bootstrap from the committed counts re-derives every number above and puts **each of the nine bootstrapped regime claims  $9.7\text{--}23.9\sigma$  beyond shot noise**; a readout-corrected fingerprint attributes the scrambled regime to predominantly coherent error (qpu\_bootstrap, qpu\_fingerprint). Controls were bought where needed: a Rigetti replication (day-drift), a same-session  $n=8$  anchor (drift), a same-window two-chip pair (generation). S5–S6 pre-committed; S7 executed (above).

## 7. Limitations (stated plainly)

(i) **No quantum advantage** is demonstrated at the  $\leq 16$ -qubit simulable scale; HAR-X (classical, linear) is the best model on this task. (ii) The RV sample ends Feb-2020 (COVID just outside); broader assets/periods are daily-proxy supporting studies (v2\_research — same negative), not 5-min RV. (iii) Task-level noise: per-layer channels (§5.4) and shot noise; combined noisy-plus-shot execution is measured in the QPU campaigns (§6). (iv)  $g$  is regularization- and configuration-dependent (qualitative gap only). (v) Superconducting  $n=8$  execution is a **characterized negative** (Fig. 3; day-scale drift exceeds shot noise, so *regime* — not point value — is the robust claim), while  $n=10/n=12$  and the newer generation land inside their limits: the wall is instance- and generation-dependent, not absolute ( $n=8$  scrambling is seed-0-instance-specific — S7; and in a 30-instance ensemble the scrambled instance is among the *sparsest*, so gate count **anti-predicts** the wall — §8). Mitigation recovery on hardware is at best marginal (ZNE  $0.223 \rightarrow 0.217$  on Rigetti; the ion chain is flat). (vi) Distinctness and full-rank entanglement are *necessary, not sufficient* (measured:  $\$3$ ,  $r \approx 0$  vs accuracy over 30 instances); whether they convert beyond the simulable frontier is open. A quantum-data probe shows the QRC natively reads nonlinear *state* functionals a linear readout cannot, but the proper baseline — **classical shadows** (Huang–Kueng–Preskill 2020) at matched budgets — **wins wherever real information is extracted**, including the shadows-hard  $\text{Tr}(\rho^3)$ : closed negatively at simulable scale (shadows\_hard\_findings.md).

## 8. Stakeholder impact, milestone plan, AI disclosure

**The durable asset is the audit instrument** — and our data shows it is *necessary*. Denser instances face a **tighter** bar ( $r = -0.40$ ,  $p = 0.03$ ), yet structure **mis-ranks the two instances we measured**: denser, tighter-bar seed-1 was signal-bearing while sparser seed-0 scrambled; and the density  $\rightarrow$  entanglement link is only suggestive ( $r = +0.34$ ,  $p = 0.07$ ). Structure does not give the answer; measurement does (instance\_ensemble\_findings.md, exploratory). Hence **trust in a quantum yes/no**: a pre-registered, vendor-neutral **workload  $\rightarrow$  QPU qualification** earned by an instrument that **falsifies its own predictions** — it split our runs **five signal-bearing, two scrambled** (Fig. 3) at  $\approx \$70\text{--}\$205$  per campaign. Target profiles only; no customers or LOIs claimed. **Desk impact** (cli.py run economics): vol-timing the GFC window **halves max drawdown ( $-61\% \rightarrow -32\%$ )** and lifts Sharpe  **$0.00 \rightarrow 0.14$**  net of costs — risk control, not alpha — but **plain HAR** captures it (CE fee  $-25\text{bp/yr}$ ): the lever is a **simple RV forecast, not quantum hardware**. **Milestone plan**: (i)–(vi) ✓ (§6). **AI disclosure**: Claude (Anthropic) assisted with code and drafting; all decisions and results are the team's own.

## References

Kornjača et al. 2024 · Zhu et al. 2025 · Ahmed, Tennie & Magri 2025 · Li et al. 2025 · Tandon et al. 2025 · Hou et al. 2025 · Čindrak et al. 2026 · Antoncich et al. 2026 · Kobayashi & Motome 2026 · Huang et al. 2021 · Huang, Kueng & Preskill 2020 (classical shadows) · Dambre et al. 2012 · Corsi 2009 · Patton 2011 · Hansen et al. 2011 · Harvey et al. 1997 · Diebold & Mariano 1995 · Bollerslev 1986 · Jaeger 2001 · Heber et al. 2009.